NATIONAL ENGINEERING INNOVATION INITIATIVE

PROJECT JAL-SHODHAK (BIO-HARVESTER)

Autonomous Amphibious Waterway Restoration Drone with In-Situ Biomass-to-Biofuel Extrusion

DISCIPLINE	FOCUS AREA	CORE INNOVATION	OUTPUT
Mechanical & Mechatronics	Canal Desiltation & Irrigation	Twin-Catamaran Extrusion Drone	Clean Canals + Biofuel Pellets

National Mission Statement: To safeguard our nation's agrarian arteries by deploying autonomous amphibious robotic harvesters that clear weed-choked irrigation canals, restore natural drainage to prevent catastrophic monsoon floods, and convert noxious aquatic biomass into decentralized clean biofuel for rural communities.

1 The National Crisis & Problem Statement

Millions of kilometers of vital irrigation canals, drainage ditches, and river networks suffer from severe biological and hydrological choke points that threaten national food security and public safety:

Core Structural Threats:

- **Invasive Weed Infestation:** Water Hyacinth (*Eichhornia crassipes*) doubles every 12 days, cutting canal water flow rate by **up to 70%** and suffocating aquatic ecology.
- **Agrarian Deprivation & Monsoon Floods:** Choked canals prevent irrigation waters from reaching tail-end farming communities during critical sowing periods, while causing violent embankment breaches and flash floods during monsoons.
- **Hazardous Conventional Remediation:** Heavy diesel earthmovers (JCBs) operate from banks, fracturing earthen lining, burning ~14 L/hr of diesel, and dumping rotting piles on roadsides that generate foul odor, disease vectors, and methane emissions.

🚨 Conventional Backhoe Dredging

Destroys canal earthen bunds, consumes massive diesel, leaves sub-surface roots intact, and creates decomposing roadside waste heaps.

💡 The Jal-Shodhak Solution

Ultralight amphibious craft operating in-channel. Solar-electric powered, cuts roots 0.8m deep, shreds and converts wet weed into solid biofuel pellets on-board.

2 System Architecture & In-Situ Conversion Pipeline

The system unites four synchronized processing modules into a continuous, self-propelled circular pipeline:

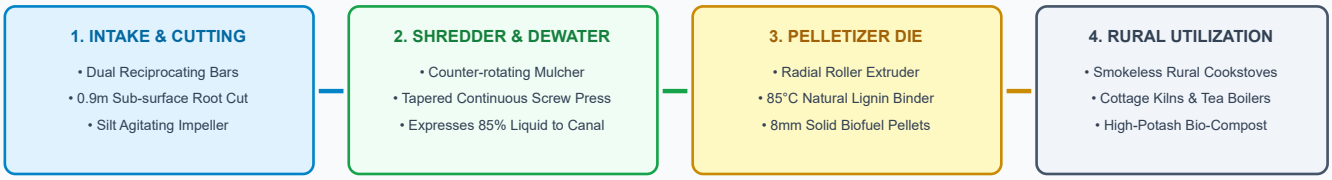


Figure 1: On-board continuous processing pipeline converting raw aquatic weeds into high-density energy pellets.

3 Mechanical & Mechatronic Engineering Subsystems

A. Amphibious Twin-Catamaran Hull & Crawlers

Built from marine-grade corrosion-resistant **Aluminum 5083-H116** and rotomolded HDPE floats. Features an ultra-shallow operational draft of **0.22 meters** to glide effortlessly over shallow weed beds and mud shoals.

- **Propulsion:** Dual 48V steerable BLDC Kort-nozzle thrusters with weed-chopping hub blades to prevent fouling.
- **Bank Traversal:** Retractable paddle-crawler tracks allow the drone to climb across canal bunds, locks, and weirs.

B. Submerged Harvester & Mulcher

Front articulated conveyor table equipped with dual reciprocating cutter bars engineered with **tungsten-carbide serrated teeth**. Submerges up to 0.9 m to cleanly slice root crowns.

- **Hexagonal Shredder:** High-torque twin shafts running at 180 RPM shred dense vegetation into 15–20 mm uniform mulch.
- **Anti-Jam Torque Monitor:** Dynamic load sensing automatically reverses cutters if nylon rope or scrap debris is detected.

C. Continuous Screw-Press Dewaterer

Water hyacinth contains 92–95% water by weight. Jal-Shodhak uses a progressive tapered compression screw with a slotted wedge-wire cylindrical screen.

- **Moisture Drop:** Reduces moisture from 94% down to under 35% in seconds.
- **Clean Effluent Return:** Filtered, oxygenated water returns to the canal, replenishing local dissolved oxygen.

D. Radial Die Extrusion Pelletizer

The compressed biomass cake enters a rotary roller die extruder. Motor cooling jacket heat (85°C) liquefies the plant's natural lignin, bonding dense 8mm solid fuel pellets without chemical additives.

- **Energy Density:** ~16.8 MJ/kg (competes directly with dry wood briquettes).
- **On-Board Bunker:** 350 kg capacity with automated offloading auger for seamless canal-bank unloading.

4 Autonomous Navigation, Sensor Fusion & Telemetry

Operates autonomously via centimeter-accurate satellite positioning and onboard Edge AI computer vision:

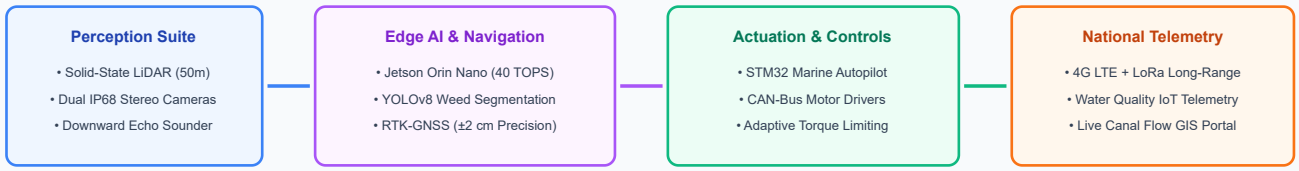


Figure 2: Mechatronics sensor-to-actuation architecture with real-time cloud GIS synchronization.

5 Technical Specifications & Bill of Materials (BOM)

Subsystem	Component Specification	Qty	Key Technical Metric	Est. Cost ($)
Hull & Chassis	Al-5083 Catamaran Hull + Rotomolded HDPE Floats	1 Unit	Displacement: 950 kg; Draft: 0.22 m	$1,850
Propulsion	Dual 48V 1.8kW BLDC Thrusters with Kort Nozzles	2 Units	Bollard Thrust: 65 kgf; Max Speed: 6 knots	$920
Cutter Bar	Reciprocating Tungsten-tipped Cutter Bar	1 Set	Cut Width: 1.6 m; Depth: 0.9 m; 1200 rpm	$780
Shredder/Press	Twin-Shaft Shredder + Tapered Screw Dewaterer	1 Unit	Torque: 420 Nm; Moisture Drop: 94% → 32%	$1,450
Pelletizer Die	Continuous Radial Die Pellet Mill (4kW BLDC Core)	1 Unit	Pellet Diam: 8mm; Throughput: 120 kg/hr dry	$1,100
Solar Canopy	High-Efficiency Monocrystalline Bifacial Panels	4x 320W	Total 1.28 kWp; MPPT Charging Controller	$640
Battery Pack	LiFePO4 48V 120Ah Deep-Cycle Power Storage	2 Packs	11.5 kWh total capacity; 3500+ lifecycle	$1,600
Compute & Vision	NVIDIA Jetson Orin Nano + IP68 Stereo Depth Cams	1 Set	YOLOv8 Edge Model for Weed / Trash Classification	$750
Navigation/IoT	Dual-Antenna RTK-GNSS + 4G LTE/LoRa Gateway	1 Set	Centimeter Positioning + Cloud Irrigation Portal	$420
TOTAL PROTOTYPE DEVELOPMENT BUDGET (Phase-1 Scale Unit)			**Ready for Field Trials**	**$9,510**

6 Comparative Advantage Over Conventional Methods

Comparison Parameter	Conventional JCB / Excavator	Imported Weed Harvester Boats	Jal-Shodhak Bio-Harvester
Capital Cost (Capex)	$55,000 – $80,000	$150,000 – $320,000	**$9,500 – $14,000** 85% Lower
Operational Energy	Diesel (~14 L/hr = $18/hr)	Diesel (~22 L/hr = $28/hr)	**Solar + LiFePO4 ($0.40/hr)** 98% Cheaper
Canal Bund Impact	Severe erosion & road collapse	Requires wide docks; damages bunds	**Zero bank impact** (Amphibious float)
Biomass Disposition	Rotting heaps on roads; toxic runoff	Dumped on barges; double transport	**In-situ pelletization** (Valuable fuel)
Carbon Footprint	Net Emitting (~38 kg CO2 / hour)	Net Emitting (~58 kg CO2 / hour)	**Net Carbon Negative** (Biofuel offset)

Circular Economy Revenue Model: A single drone clearing 1.2 hectares per 8-hour shift produces approximately **1,800 kg of dense dry biofuel pellets daily**. Sold to local rural brick kilns or tea factories at $0.08/kg, the drone generates **~$144/day in clean green revenue**, fully amortizing its capital construction cost in less than 4 months of operation!

7 12-Month Prototype & National Deployment Roadmap

Quarter 1: CAD & Lab Bench Prototyping

- Hydrodynamic simulation (ANSYS Fluent) of catamaran hull under weed drag.
- Fabrication of scale screw-press dewaterer; torque and dewatering test bench.
- Dataset collection and training of YOLOv8 weed-segmentation model.

Quarter 2: Full-Scale Fabrication

- Welding and CNC fabrication of AI-5083 catamaran hull and conveyor structure.
- Integration of 48V powertrain, solar canopy, and LiFePO4 battery management.
- Static buoyancy, balance, and emergency bilge pump validation in tank test.

Quarter 3: Irrigation Canal Field Trials

- Deployment in a 5 km heavily choked agricultural canal pilot stretch.
- Validation of automated RTK-GNSS waypoint navigation and collision avoidance.
- Chemical calorific testing (bomb calorimeter) of extruded biofuel pellets.

Quarter 4: Autonomous Fleet Swarm & Handover

- Implementation of multi-drone swarm coordination for broad lake clearing.
- Creation of open-source maintenance manual and training for rural youth cooperatives.
- Formal presentation to National Irrigation and Water Resources Ministries.

8 Direct National Impact & Sovereign Benefits

🌾 Food Security & Irrigation Flow

Guarantees full designed hydraulic flow through canal networks, ensuring vital irrigation water reaches vulnerable downstream smallholder farmers on schedule during dry sowing seasons.

🛡️ Flood Mitigation & Disaster Resilience

Pre-monsoon canal desiltation and weed clearing prevents backflow surges and structural embankment breaches that cause catastrophic loss of life and rural livestock annually.

⚡ Rural Clean Fuel Independence

Supplies women and rural households with clean, smokeless biomass pellets, replacing dangerous indoor firewood burning and reducing acute respiratory illnesses.

IN Atmanirbhar Technological Sovereignty

Replaces multi-crore imported diesel dredger boats with indigenous, open mechatronic technology built, serviced, and operated locally by national engineering talent.

Conclusion & Call to Action: *Project Jal-Shodhak* demonstrates how mechanical engineering, robotics, and circular economy design can directly solve one of the nation's most persistent ecological and agrarian crises. By turning an environmental hazard into a source of clean energy and water security, this project serves both the land and its people.

Project Proposal Submitted By: Engineering Research & Innovation Group
Department: Mechanical, Mechatronics & Autonomous Systems

Verification Status: Ready for Prototype Stage
Documentation: Complete Technical Blueprint (PDF)